Sumanth **Doddapaneni**

Research Scientist | Google DeepMind

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Education

(On Leave) Jan 2022 | **Indian Institute of Technology (IIT), Madras** — **Chennai, India**
Ph.D, Computer Science & Engineering
Advisors: Mitesh M. Khapra, Anoop Kunchukuttan

May 2020 Aug 2016 | **Indian Institute of Information Technology (IIIT), Sri City** — **Sri City, India**
B.Tech., Electronics & Communications Engineering

Experience

Present Sep 2026 | **Google DeepMind** — **Bangalore, India**
Research Scientist
- Working on improving agents for {lets keep it a mystery for now!}

Sep 2026 Apr 2025 | **Sarvam AI** — **Bangalore, India**
ML Researcher
- Trained 3B Hybrid-Mamba, 30B and 105B MoE models with end-to-end ownership of the training stack, including architectural ablations, distributed infrastructure management, and real-time automated evals.
- Led the end-to-end development of a SOTA multilingual translation system, Sarvam Translate, for 22 Indian languages. Addressed the "structure gap" by enabling translation of HTML, LaTeX, and Code for 15+ languages. Model outperforms Llama-3.1-405B and Gemma3-27B in human evaluations.

Mar 2024 Jun 2023 | **Google Research** — **Mountain View & Bangalore**
Research Intern | Hosts: Krishna Sayana, Nitish Gupta, Partha Talukdar
- [p1] Investigated the capability of LLMs to infer user preferences from behavioral history, revealing critical insights into explicit vs. implicit user signals compared to traditional recommender systems. [EACL'24]
- [p2] Proposed QAPGEN, a novel inference-time planning algorithm that leverages English reasoning to enhance long-form generation in 16 languages. Demonstrated that QAPGEN significantly outperforms vanilla decoding and proprietary baselines in human evaluations.

Apr 2025 Oct 2020 | **AI4Bharat, IIT Madras** — **Chennai, India**
PhD Researcher | Advisors: Mitesh M. Khapra, Anoop Kunchukuttan, Pratyush Kumar
- Developed FBI [EMNLP'24] and CIA [ACL'25] frameworks for robust LLM assessment. FBI exposed a 50% failure rate in existing evaluators via targeted perturbations, while CIA's Hercule model outperformed proprietary models like GPT-4 in human alignment for low-resource languages.
- Developed SoTA models for Machine Translation (IndicTrans v1 [TACL'22] & v2 [TMLR'23]) and Automatic Speech Recognition (IndicWav2Vec [AAAI'22]) for Indian languages beating all the publicly available models and on par with proprietary APIs.
- Led development of data, models, and evaluation benchmarks for 22 Indian languages, producing the then-largest multilingual corpus and state-of-the-art encoder models across languages. [ACL'23 & ACL'24]

Select Publications Full List

P=Preprints, C=Conference, W=Workshop, J=Journal, *=Equal Contribution

[P] **Sarvam Sovereign Models** [Blog][Playground] [Model]
Sumanth Doddapaneni with team at Sarvam AI

[P] **Sarvam Translate** [Blog][Playground] [Model]
Sumanth Doddapaneni with team at Sarvam AI

[C] **Cross-Lingual Auto Evaluation for Assessing Multilingual LLMs** [][Code] [Models]
Sumanth Doddapaneni*, Mohammed Safi Ur Rahman Khan*, *et al.*
The 63rd Annual Meeting of the Association for Computational Linguistics **[ACL 2025]**

[C] [Outstanding Paper] **Finding Blind Spots in Evaluator LLMs with Interpretable Checklists** [🔗][Code][Data]
Sumanth Doddapaneni*, Mohammed Safi Ur Rahman Khan*, Sshubam Verma, Mitesh M. Khapra
The 2024 Conference on Empirical Methods in Natural Language Processing [**EMNLP 2024**]

[C] [Outstanding Paper] **IndicLLMSuite: A Blueprint for Creating Pre-training and Fine-Tuning Datasets for Indian Languages** [🔗][Code]
Mohammed Safi Ur Rahman Khan, Priyam Mehta, ..., Sumanth Doddapaneni, *et al.*
62nd Annual Meeting of the Association for Computational Linguistics, Bangkok, Thailand [**ACL 2024**]

[J] **IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages** [🔗][Code]
Jay Gala, Pranjal A Chitale, Raghavan AK, Sumanth Doddapaneni, *et al.*
Transactions on Machine Learning Research [**TMLR 2023**]

[J] **Samanantar: The Largest Publicly Available Parallel Corpora Collection for 11 Indic Languages** [🔗][Code]
Gowtham Ramesh*, Sumanth Doddapaneni*, *et al.*
Transactions of the Association for Computational Linguistics [**TACL 2022**]

[C] **Towards Leaving No Indic Language Behind: Building Monolingual Corpora, Benchmark and Models for Indic Languages** [🔗][Code]
Sumanth Doddapaneni, Rahul Aralikatte, Gowtham Ramesh, Shreya Goyal,
Mitesh M. Khapra, Anoop Kunchukuttan, Pratyush Kumar
61st Annual Meeting of the Association for Computational Linguistics, Toronto, Canada [**ACL 2023**]

Honours and Grants

Outstanding Paper Award @ EMNLP 2024 Finding Blind Spots in Evaluator LLMs with Interpretable Checklists

Outstanding Paper Award @ ACL 2024 IndicLLMSuite: A Blueprint for Creating Pre-training and Fine-Tuning Datasets for Indian Languages

Google PhD Fellowship 2023 (List of Recipients)

Select PhD Research Projects

Auto Evaluation with LLMs [Code][Models] Apr'24 - Mar'25
Advisors: Dr. Mitesh M. Khapra

- Analyzed failure modes in **automated evaluators** using systematic perturbations, identifying significant shortcomings in existing evaluation methods.
- Developed the **first cross-lingual auto-evaluator** and released a **multilingual benchmark**, enabling comprehensive evaluation of multilingual LLMs.
- Published results at **EMNLP 2024** [Outstanding Paper Award] and ACL 2025.

Machine Translation [Try the model][Code] Feb'21 - May'23
Advisors: Dr. Mitesh M. Khapra, Dr. Anoop Kunchukuttan, Dr. Pratyush Kumar

- Developed state-of-the-art translation models for Indian languages, releasing two versions—**IndicTrans v1** and **v2**—over multiple years to enhance multilingual performance.
- Built the largest bilingual corpus, **Samanantar** ($\sim$ 50M sentence pairs), spanning 11 Indian languages for NMT training. This effort was further expanded in the v2 release with **BPCC** ($\sim$ 230M sentence pairs) across 22 languages.
- Published papers at **TACL** and **TMLR**.

Language Modeling [🔗] Dec'21 - Feb'24
Advisors: Dr. Mitesh M. Khapra, Dr. Anoop Kunchukuttan, Dr. Pratyush Kumar

- Developed the largest monolingual corpus, **IndicCorp v2**, followed by an even larger version, **Sangraha**, in the subsequent release.
- Built language models, including **IndicBERT v2** for masked language modeling and **VartaT5** for text generation, and created comprehensive multilingual benchmarks for Indian languages, **IndicXTREME**.
- Published multiple papers at **ACL 2023** and **ACL 2024**, with IndicLLMSuite receiving the [Outstanding Paper Award].